

Ex.no:9	MINI PROJECT – SPORTS DATA ANALYTICS

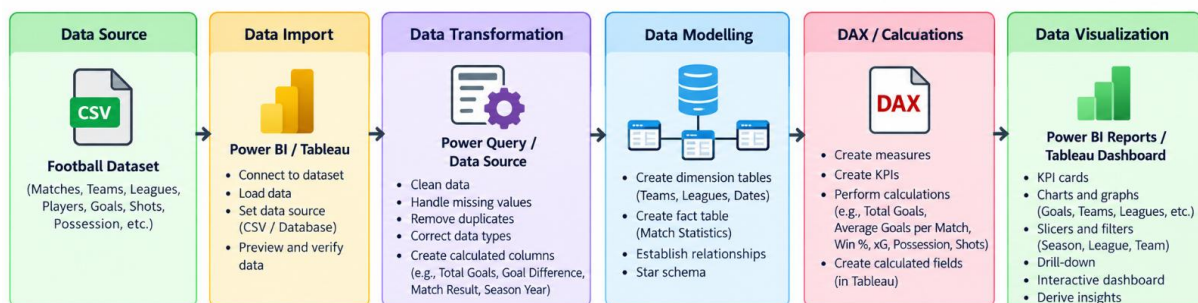
Aim:

To develop an interactive Sports Data Analytics Dashboard using Power BI and Tableau to analyze football match information, teams, leagues, match results, goals, expected goals (xG), possession, shots, passes, fouls, cards, and other match performance statistics.

Algorithm:

1. Import the Football Match Dataset into Power BI and Tableau.
2. Clean and transform the data using Power Query in Power BI and Data Source in Tableau.
3. Check and correct data types and remove unwanted, duplicate, or inconsistent records.
4. Create required calculated fields such as Total Goals, Match Result, Goal Difference, Total Shots, and Average xG.
5. Create the required data model and relationships in Power BI using fact and dimension tables.
6. Create required DAX measures and KPIs such as Total Matches, Total Goals, Average Goals per Match, Home Win %, and Away Win %.
7. Create calculated fields and required measures in Tableau.
8. Create charts for team performance, league performance, goals, match results, xG, possession, shots, passing, fouls, and cards.
9. Add cards, KPIs, slicers, filters, and drill-downs in Power BI.
10. Add filters, dashboards, parameters, and interactive visualizations in Tableau.
11. Design and format the Power BI and Tableau dashboards for clear presentation of football performance.
12. Analyze the visualizations and identify important football performance trends and insights.
13. Present the final results through the interactive Power BI and Tableau dashboards.
14. End.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
import numpy as np
from sqlalchemy import create_engine, text

# Load dataset
df = pd.read_csv("Football(1).csv")

# Create a copy for processing
football = df.copy()

# Display original shape
print("Original Shape:", football.shape)

# Remove duplicate records
football.drop_duplicates(inplace=True)

# Handle missing values
football["Country"] = football["Country"].fillna("Unknown")
football["League"] = football["League"].fillna("Unknown")
football["home_team"] = football["home_team"].fillna("Unknown")
football["away_team"] = football["away_team"].fillna("Unknown")
football["referee"] = football["referee"].fillna("Unknown")
football["venue"] = football["venue"].fillna("Unknown")

# Convert numeric columns
numeric_columns = [
    "home_score",
    "away_score",
    "expected_goals_xg_home",
    "expected_goals_xg_host",
```

```
"Ball_Possession_Home",
"Ball_Possession_Host",
"Goal_Attempts_Home",
"Goal_Attempts_Host",
"Shots_on_Goal_Home",
"Shots_on_Goal_Host",
"Shots_off_Goal_Home",
"Shots_off_Goal_Host",
"Corner_Kicks_Home",
"Corner_Kicks_Host",
"Fouls_Home",
"Fouls_Host",
"Yellow_Cards_Home",
"Yellow_Cards_Host",
"Red_Cards_Home",
"Red_Cards_Host",
"Total_Passes_Home",
"Total_Passes_Host",
"Completed_Passes_Home",
"Completed_Passes_Host",
"Tackles_Home",
"Tackles_Host",
"Interceptions_Home",
"Interceptions_Host",
"Attacks_Home",
"Attacks_Host",
"Dangerous_Attacks_Home",
"Dangerous_Attacks_Host"
]

for col in numeric_columns:
    football[col] = pd.to_numeric(
```

```
    football[col],  
    errors="coerce"  
)
```

```
# Fill missing numerical values with 0
```

```
football[numeric_columns] = football[numeric_columns].fillna(0)
```

```
# Create Total Goals
```

```
football["total_goals"] = (  
    football["home_score"] +  
    football["away_score"]  
)
```

```
# Create Goal Difference
```

```
football["goal_difference"] = abs(  
    football["home_score"] -  
    football["away_score"]  
)
```

```
# Create Match Result
```

```
football["match_result"] = np.select(  
    [  
        football["home_score"] > football["away_score"],  
        football["home_score"] == football["away_score"],  
        football["home_score"] < football["away_score"]  
    ],  
    [  
        "Home Win",  
        "Draw",  
        "Away Win"  
    ],  
    default="Unknown"
```

)

Create Total Shots

```
football["total_shots"] = (  
    football["Goal_Attempts_Home"] +  
    football["Goal_Attempts_Host"]  
)
```

Create Total Shots on Goal

```
football["total_shots_on_goal"] = (  
    football["Shots_on_Goal_Home"] +  
    football["Shots_on_Goal_Host"]  
)
```

Create Total Corners

```
football["total_corners"] = (  
    football["Corner_Kicks_Home"] +  
    football["Corner_Kicks_Host"]  
)
```

Create Total Fouls

```
football["total_fouls"] = (  
    football["Fouls_Home"] +  
    football["Fouls_Host"]  
)
```

Create Total Yellow Cards

```
football["total_yellow_cards"] = (  
    football["Yellow_Cards_Home"] +  
    football["Yellow_Cards_Host"]  
)
```

```
# Create Total Red Cards
```

```
football["total_red_cards"] = (  
    football["Red_Cards_Home"] +  
    football["Red_Cards_Host"]  
)
```

```
# Create Total Passes
```

```
football["total_passes"] = (  
    football["Total_Passes_Home"] +  
    football["Total_Passes_Host"]  
)
```

```
# Create Total Completed Passes
```

```
football["total_completed_passes"] = (  
    football["Completed_Passes_Home"] +  
    football["Completed_Passes_Host"]  
)
```

```
# Create Average Possession
```

```
football["average_possession"] = (  
    football["Ball_Possession_Home"] +  
    football["Ball_Possession_Host"]  
) / 2
```

```
# Create Total Expected Goals
```

```
football["total_xg"] = (  
    football["expected_goals_xg_home"] +  
    football["expected_goals_xg_host"]  
)
```

```
# Create xG Difference
```

```
football["xg_difference"] = abs(
```

```
    football["expected_goals_xg_home"] -  
    football["expected_goals_xg_host"]  
)
```

```
# Extract day from Date_day  
football["match_day"] = pd.to_numeric(  
    football["Date_day"]  
    .astype(str)  
    .str.split(".")  
    .str[0],  
    errors="coerce"  
)
```

```
# Standardize column names  
football.columns = (  
    football.columns  
    .str.lower()  
    .str.strip()  
    .str.replace(" ", "_")  
    .str.replace("(", "", regex=False)  
    .str.replace(")", "", regex=False)  
    .str.replace("/", "_", regex=False)  
)
```

```
# Display original and cleaned shape  
print("Original Shape:", df.shape)  
print("Cleaned Shape:", football.shape)
```

```
football.head()
```

2. MySQL – Database and Data Loading

MySQL Command Prompt

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL

```
CREATE DATABASE IF NOT EXISTS football_analytics;
```

```
USE football_analytics;
```

Python / Jupyter

```
from sqlalchemy import create_engine, text
```

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(  
    f'mysql+pymysql://{username}:{password}@{host}:{port}/football_analytics'  
)
```

```
# Load cleaned data into MySQL
```

```
football.to_sql(  
    "football_matches",  
    con=db_engine,  
    if_exists="replace",  
    index=False,  
    chunksize=5000  
)
```

```
print("Football data loaded successfully into MySQL.")
```

Verify

```
pd.read_sql(  
    text("SELECT * FROM football_matches"),  
    db_engine
```

```
"SELECT COUNT(*) AS total_rows FROM football_matches",  
db_engine  
)
```

You can also verify the top leagues:

```
pd.read_sql(  
    """  
    SELECT  
        league,  
        COUNT(*) AS total_matches  
    FROM football_matches  
    GROUP BY league  
    ORDER BY total_matches DESC  
    LIMIT 10  
    """,  
    db_engine  
)
```

3. Power BI – DAX Measures

Total Matches

Total Matches =
COUNTROWS(FactMatches)

Total Goals

Total Goals =
SUM(FactMatches[total_goals])

Average Goals per Match

Average Goals per Match =
AVERAGE(FactMatches[total_goals])

Home Wins

Home Wins =
CALCULATE(
 [Total Matches],
 FactMatches[match_result] = "Home Win"

)

Away Wins

Away Wins =

CALCULATE(

[Total Matches],

FactMatches[match_result] = "Away Win"

)

Draws

Draws =

CALCULATE(

[Total Matches],

FactMatches[match_result] = "Draw"

)

Home Win Rate

Home Win Rate =

DIVIDE(

[Home Wins],

[Total Matches]

)

Away Win Rate

Away Win Rate =

DIVIDE(

[Away Wins],

[Total Matches]

)

Draw Rate

Draw Rate =

DIVIDE(

[Draws],

[Total Matches]

)

Total Expected Goals

Total xG =

SUM(FactMatches[total_xg])

Average xG

Average xG =

AVERAGE(FactMatches[total_xg])

Total Shots

Total Shots =

SUM(FactMatches[total_shots])

Shots on Goal

Total Shots on Goal =

SUM(FactMatches[total_shots_on_goal])

Total Corners

Total Corners =

SUM(FactMatches[total_corners])

Total Fouls

Total Fouls =

SUM(FactMatches[total_fouls])

Total Yellow Cards

Total Yellow Cards =

SUM(FactMatches[total_yellow_cards])

Total Red Cards

Total Red Cards =

SUM(FactMatches[total_red_cards])

Total Passes

Total Passes =

SUM(FactMatches[total_passes])

Completed Passes

Completed Passes =

SUM(FactMatches[total_completed_passes])

Pass Completion Rate

Pass Completion Rate =

DIVIDE(

```
[Completed Passes],  
[Total Passes]  
)  
Average Possession  
Average Possession =  
AVERAGE(FactMatches[average_possession])
```

```
Average Goal Difference  
Average Goal Difference =  
AVERAGE(FactMatches[goal_difference])
```

4. Tableau – Calculated Fields

```
Total Matches  
COUNT([Home Team])
```

```
Total Goals  
SUM([Total Goals])
```

```
Average Goals per Match  
AVG([Total Goals])
```

```
Home Wins  
SUM(  
    IF [Match Result] = "Home Win"  
    THEN 1  
    ELSE 0  
    END  
)
```

```
Away Wins  
SUM(  
    IF [Match Result] = "Away Win"  
    THEN 1  
    ELSE 0  
    END  
)
```

```
Draws  
SUM(  
    IF [Match Result] = "Draw"  
    THEN 1  
    ELSE 0  
    END  
)
```

```
    IF [Match Result] = "Draw"
    THEN 1
    ELSE 0
    END
)
```

Home Win Rate

```
SUM(
    IF [Match Result] = "Home Win"
    THEN 1
    ELSE 0
    END
)
/
COUNT([Home Team])
```

Away Win Rate

```
SUM(
    IF [Match Result] = "Away Win"
    THEN 1
    ELSE 0
    END
)
/
COUNT([Home Team])
```

Draw Rate

```
SUM(
    IF [Match Result] = "Draw"
    THEN 1
    ELSE 0
    END
)
/
COUNT([Home Team])
```

Total xG

SUM([Total Xg])

Average xG

AVG([Total Xg])

Total Shots

SUM([Total Shots])

Shots on Goal

SUM([Total Shots On Goal])

Total Corners

SUM([Total Corners])

Total Fouls

SUM([Total Fouls])

Total Yellow Cards

SUM([Total Yellow Cards])

Total Red Cards

SUM([Total Red Cards])

Average Possession

AVG([Average Possession])

Pass Completion Rate

SUM([Total Completed Passes])

/

SUM([Total Passes])

Match Result

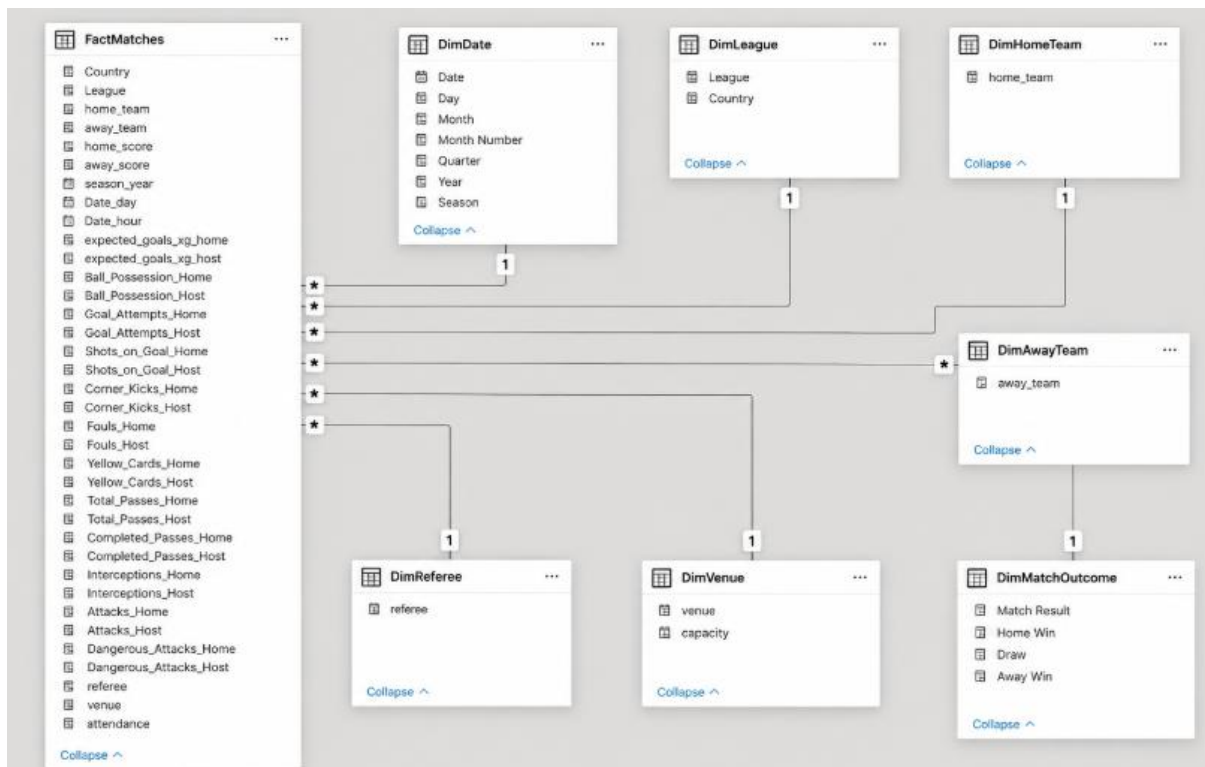
IF [Home Score] > [Away Score] THEN "Home Win"

ELSEIF [Home Score] = [Away Score] THEN "Draw"

ELSE "Away Win"

END

Output:

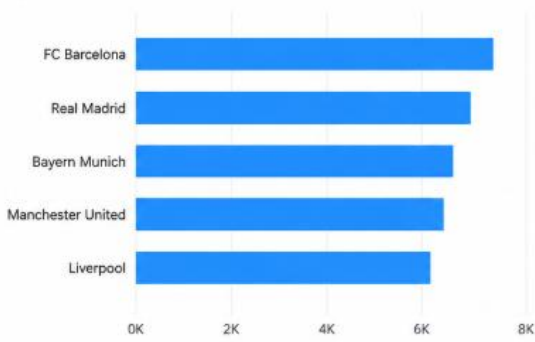


Total Goals and Average Goals per Match

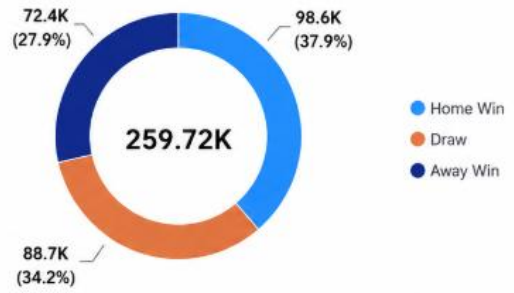
by Date



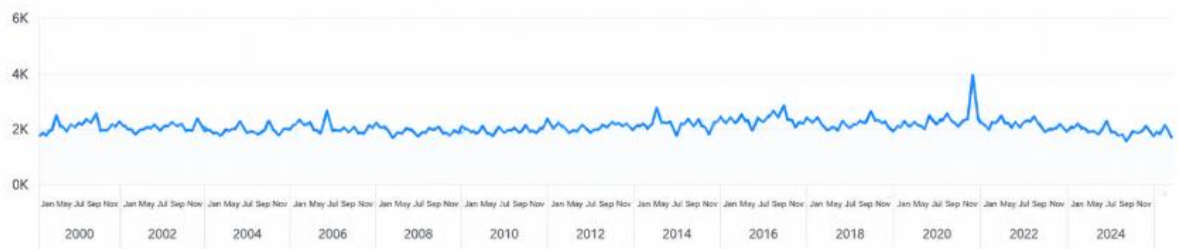
Total Goals
by Team



Total Goals
by Match Result



Total Goals
by Year and Month



Total Matches
100.00K

Average Goals
per Match
2.60

Total Goals
259.72K

Average Shots
per Match
12.35

Year
2000 2024

Country
☐ England
☐ France
☐ Germany
☐ Italy
☐ Spain
☐ Turkey

Match Result
☐ Home Win
☐ Draw
☐ Away Win

League

All

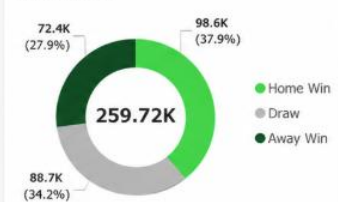
Team

All

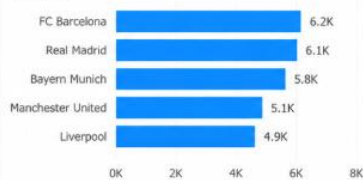
Total Goals and Average Goals per Match
by Date



Total Goals
by Match Result



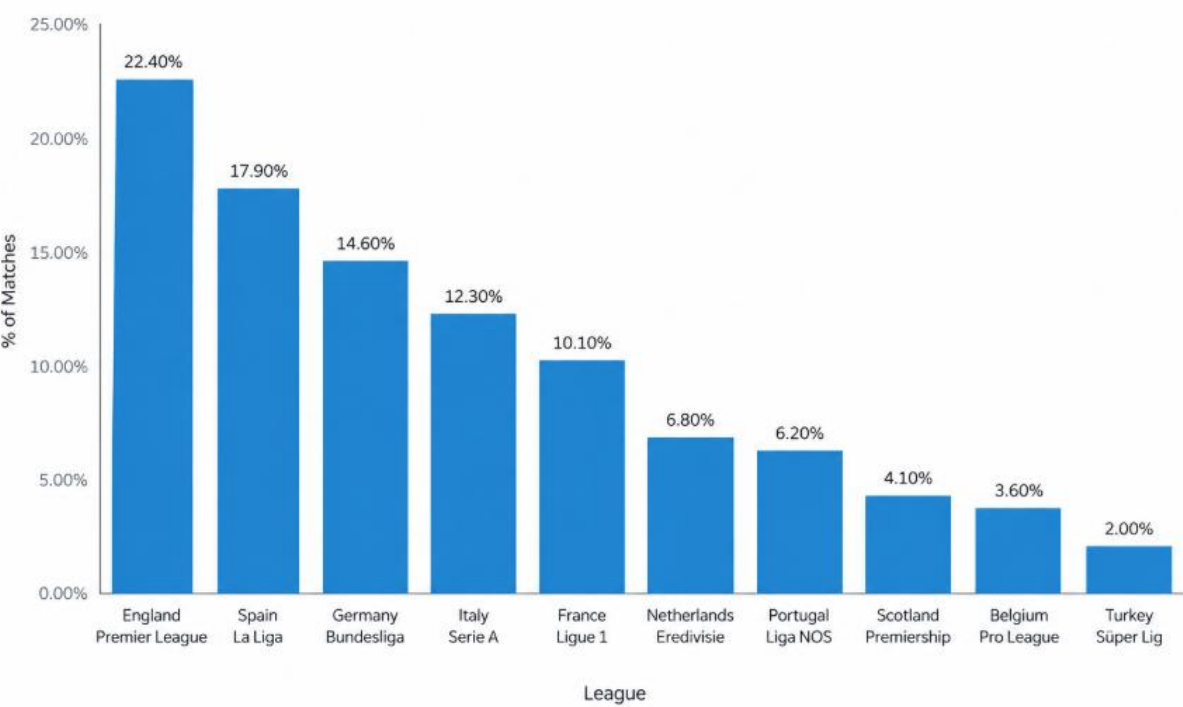
Total Goals
by Team



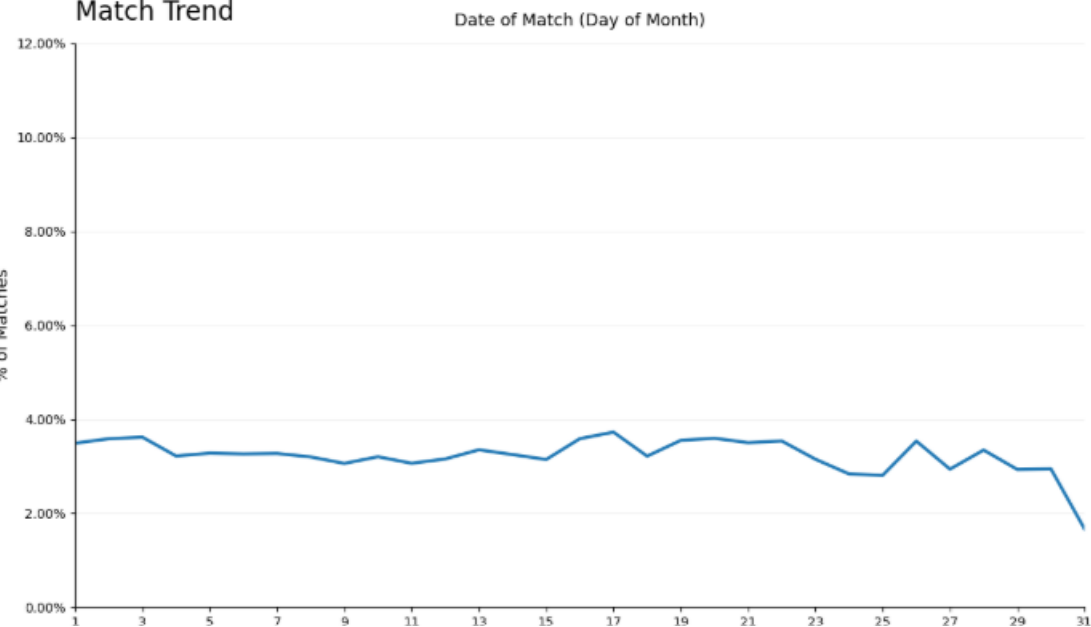
Total Goals
by Year and Month

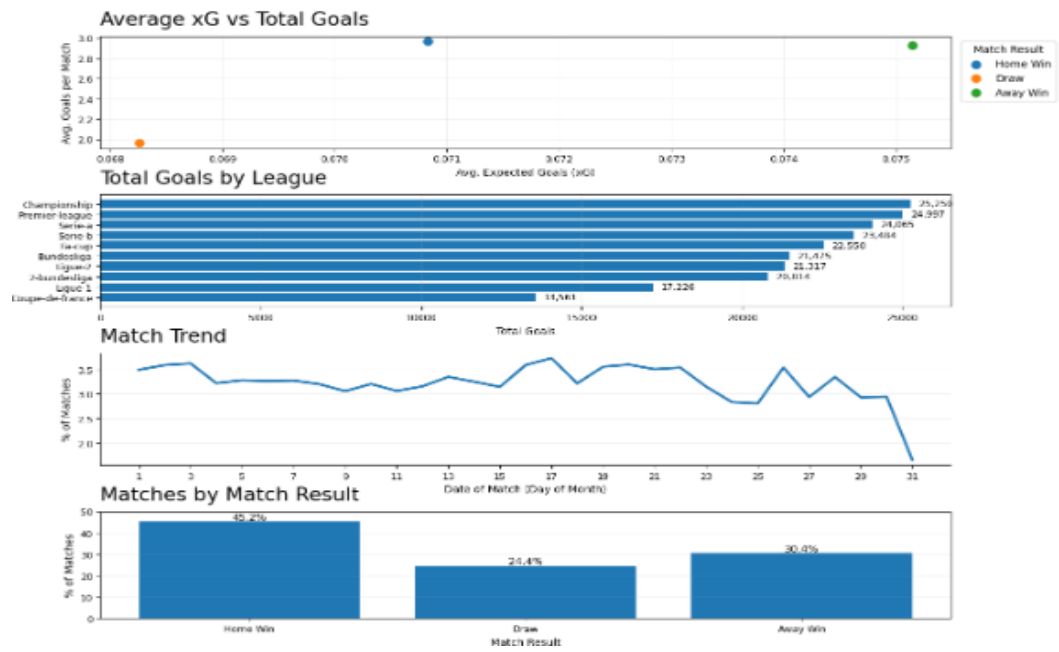


Matches by League



Match Trend





Result: